

The Immunity Factor from Contact Surveys: Where Epidemics Turn Over

Peter Cotton*

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Abstract

Many COVID-19 waves reached their peak of incidence when estimated cumulative infections were well below the threshold $1 - 1/R$ implied by their initial growth under a homogeneous model. Because people with many contacts are infected sooner, the effective reproduction number falls faster than the susceptible share S , approximately as RS^λ , where the immunity factor λ is one in a homogeneous population; the peak then comes at about $1 - R^{-1/\lambda}$. In repeated UK contact surveys, a model of persistent and changing activity predicts $\lambda = 3.4\text{--}3.7$ when daily contacts are capped at 20 and 4.6–6.8 at caps of 50 and 100, without fitting to epidemic outcomes. Winter 2020–21 US county waves imply about 3.5 when growth is estimated from deaths and the infection fatality rate is 0.7%, and 2.4–5.3 for rates from 0.5% to 1%.

1 Introduction

In 341 US county epidemic waves of winter 2020–21, the median wave reached its peak of incidence when estimated cumulative infections were 16% of the population. Under a homogeneous model, the growth at the start of those waves implies a median threshold of 30%. Infections here are estimated from deaths at a fatality rate of 0.7%, and growth is converted to a reproduction number with a generation interval of mean 5.5 days. We call the peak of incidence the *turnover*.

In the spring of 2020 case counts doubled in under a week, which a homogeneous model reads as a reproduction number near 2.5 and a threshold of 60%. Cotton (2020a) called the gap between such thresholds and the observed peaks the herd-immunity paradox and argued that it follows from variation in transmission across people, places and time: early growth is dominated by the fastest transmitters, and the turnover by a harmonic mean (Cotton, 2020b).

People with many contacts are infected sooner, and while infectious they generate more exposure, so each infection removes more than its share of future transmission. Britton et al. (2020) and Gomes et al. (2022) made this quantitative for persistent differences in activity or susceptibility. Tkachenko et al. (2021) added activity that changes over time and showed that the resulting suppression is partly transient.

The effect is summarized by the *immunity factor* λ , the rate at which the effective reproduction number falls relative to the susceptible share. We estimate it in two ways that share no data. The first uses repeated contact surveys and a model of persistent and changing activity, without

*peter.cotton@microprediction.com. Code, data sources and a dated log of results: github.com/microprediction/epidemiology; summaries at epidemiology.microprediction.org.

fitting to epidemic outcomes. The second backs a value out of the peaks of US county waves under an assumed fatality rate. The two agree at about 3.5 for a 20-contact cap and a 0.7% fatality rate, and both depend on those choices.

Where contacts were cut by lockdowns or voluntary distancing, peaks came with 1–5% infected. The value of λ implied by such a peak absorbs the drop in contacts and does not measure immunity.

2 The turnover rule

Unequal activity. Let S be the susceptible share. In a homogeneous population the effective reproduction number is $R_e = R S$. For fixed gamma-distributed activity the power law below is exact, as shown next. In other models we define λ by the initial fall of R_e and use the power law as an approximation to predict the turnover. When people differ in activity, write

$$R_e = R S^\lambda. \quad (1)$$

Early on, $R_e \approx R(1 - \lambda(1 - S))$, so each percentage point of infection lowers R_e by λ percentage points of R . Setting $R_e = 1$ gives the infected share at the turnover,

$$a^\star = 1 - R^{-1/\lambda}. \quad (2)$$

The textbook threshold is the case $\lambda = 1$. Equivalently, the textbook formula applies with the adjusted reproduction number $R^{1/\lambda}$.

Here R is the initial effective reproduction number of the model. When R is inferred from growth it also depends on the assumed generation interval, and in a model with changing activity the two need not coincide.

Gamma activity. Let activity a have a gamma distribution with mean one and squared coefficient of variation $\text{CV}^2 = 1/k$, and let it scale both the chance of infection and the number of people infected. After cumulative force of infection Z , the susceptible share is $S = (1 + Z/k)^{-k}$ and $R_e \propto \mathbb{E}[a^2 e^{-aZ}] \propto (1 + Z/k)^{-k-2}$. Hence (1) holds exactly with

$$\lambda = 1 + 2 \text{CV}^2, \quad (3)$$

as in Gomes et al. (2022).

Age structure. Let M_{ij} be the mean number of daily contacts of a person in age group i with people in group j , and n_i the population shares. Infected fractions grow along the leading right eigenvector v of M ; let u be the left eigenvector. Depleting susceptibles along v lowers the leading eigenvalue at the rate

$$\lambda_{\text{age}} = \frac{\sum_i u_i v_i^2 / \sum_i u_i v_i}{\sum_i n_i v_i}. \quad (4)$$

The numerator measures how strongly early infections sit in the groups that drive transmission; the denominator converts to the overall infected share. The same formula applies to any finite type structure, and we use it for age crossed with activity.

Accuracy of the rule. Equation (4) is a first-order slope, so (2) is an approximation away from the gamma case. In the joint age and activity models of Section 5, solved numerically to the peak, the rule understates the infected share at the turnover by 0.6 to 1.2 percentage points at $R = 1.3$, 1.4 to 2.9 points at $R = 1.6$, and 3.6 to 7.0 points at $R = 2.5$.

3 Activity that changes over time

A solvable model. Let each person's activity take two values with long-run shares p_1, p_2 , mean one and variance σ^2 , so that $(a - 1)^2 = c(a - 1) + \sigma^2$ for a constant c that sets the skew. Activity switches on a Markov clock with generator Q and relaxes at rate κ . A person with activity a is infected at rate $\beta a \Phi$, where $\Phi = \sum_k a_k I_k$, and infectious people recover at rate γ .

Proposition 1 (Growth, reproduction number and immunity factor). *The early growth rate r and the basic reproduction number solve*

$$1 = \beta \left(\frac{1}{r + \gamma} + \frac{\sigma^2}{r + \gamma + \kappa} \right), \quad R_0 = \beta \left(\frac{1}{\gamma} + \frac{\sigma^2}{\gamma + \kappa} \right). \quad (5)$$

During growth, with $\rho = r/(r + \kappa)$, the immunity factor is

$$\lambda = \frac{(1 + \sigma^2 \rho)/\gamma + \sigma^2 (1 + (1 + c)\rho)/(\gamma + \kappa)}{1/\gamma + \sigma^2/(\gamma + \kappa)}. \quad (6)$$

The proof is in Appendix A. The resolvent $(r + \gamma - Q^\top)^{-1}$ acts on only two directions, the stationary law p and the deviation $(a - 1) \circ p$, which decays at rate κ . People with more contacts are infected sooner and, while infectious, generate more exposure. The second term in (5) is the persistence of an individual's activity over the infectious period: it contributes while a busy phase outlasts the infection.

Limits. With $\kappa = 0$, (6) is $\langle a^3 \rangle / \langle a^2 \rangle$, the value for persistent heterogeneity. As $\kappa \rightarrow \infty$ it tends to one. For fast switching,

$$\lambda - 1 \approx r K_a, \quad K_a = \frac{\sigma^2}{\kappa}, \quad (7)$$

where K_a is the Green–Kubo number of activity, the integral of its autocovariance. A fast epidemic outruns the reshuffling of activity and uses up the busy phase before it is refilled.

Effective variance. For the prediction in Section 5 we combine a persistent variance σ_p^2 with a switching variance σ_s^2 of correlation time τ through

$$\text{CV}_{\text{eff}}^2 = \sigma_p^2 + \frac{r}{r + 1/\tau} \sigma_s^2, \quad (8)$$

the persistent part acting in full and the switching part in proportion to the share of a busy phase that survives one growth time. This interpolates between the limits of Proposition 1 and is an approximation.

4 Mixtures of places

Cotton (2020b) reaches the same conclusion from a population made of many places. To isolate the effect of heterogeneous local rates, consider independent places with equally seeded epidemics, each with transmission rate $\beta = \bar{\beta} \eta$ where η has mean one across places. Early growth of the total reads a transmission rate $\beta_g = t^{-1} \log \mathbb{E} e^{t\bar{\beta}\eta} = G \bar{\beta}$ with $G > 1$. The susceptible share at the turnover, averaged over places, is $\mathbb{E}[\gamma/\beta] = J/\bar{R}$ with $J = \mathbb{E}[1/\eta] \geq 1$. Together,

$$s^* \approx \frac{JG}{R_g}, \quad (9)$$

larger than the textbook $1/R_g$ by two convexity adjustments. When each place’s rate switches between $\bar{\beta} \pm \delta$ at rate ν , the total grows at the top eigenvalue $\bar{\beta} - \gamma - \nu + \sqrt{\nu^2 + \delta^2}$, which is $\bar{\beta} - \gamma + \delta^2/(2\nu)$ to first order: the growth convexity is again a Green–Kubo number.

5 The immunity factor from contact surveys

One-day diaries. The POLYMOD diaries (Mossong et al., 2008) record every contact of 7,290 people in eight European countries on one day. Within age groups the squared coefficient of variation of the number of contacts is 0.53 for ages 0–19, 0.63 for 20–64 and 0.77 for 65 and over. Combining these with the age matrices of Prem et al. (2021) through (4), with activity held fixed, gives $\lambda = 2.70$ for the United States, 2.77 for Sweden, 2.66 for Spain, 3.02 for the United Kingdom and 2.30 for Brazil. Age alone gives 1.3–1.7.

A one-day diary cannot tell a busy person from a busy day, so these values mix lasting and passing variation.

A two-year panel. The CoMix panel (Jarvis et al., 2020) surveyed the same people in the United Kingdom from March 2020 to March 2022. We use 21,158 adults aged 18–69 seen at least three times, 160,869 survey days in all. Activity is the number of contacts on the diary day divided by the mean for the same week and age group. This removes the common effect of lockdowns and seasons on each age group, though not effects that differ between people. The covariance of a person’s activity between surveys ℓ days apart is fitted as

$$C(\ell) = \sigma_p^2 + \sigma_s^2 e^{-\ell/\tau} \quad (\ell \geq 7), \quad C(0) = \sigma_p^2 + \sigma_s^2 + \sigma_n^2, \quad (10)$$

with a persistent part, a switching part of correlation time τ , and a residual σ_n^2 that repeat observations at lags of a week or more cannot resolve.

contacts capped at	one-day CV ²	persistent σ_p^2	switching σ_s^2	τ (days)	residual σ_n^2
20	1.82	0.54 (30%)	0.23 (13%)	36	1.05 (57%)
50	3.90	1.06 (27%)	0.51 (13%)	28	2.34 (60%)
100	6.54	1.69 (26%)	0.88 (14%)	23	3.97 (61%)

Table 1: Decomposition of adult activity in the CoMix UK panel.

Under this covariance model, 26–30% of the variance of capped daily contacts is persistent and 13–14% belongs to a component with a correlation time of 23 to 36 days. The remaining

57–61% is unresolved by repeat observations at lags of a week or more. It includes day-to-day variation, and it may also include genuine correlations lasting less than a week, which could matter over an infectious period. The shares are similar under the three caps, but the levels are not.

Prediction. Inserting (8) into the joint age and activity structure with the UK matrix gives the immunity factors in Table 2, for growth rates from 0.02 to 0.2 per day. The residual is left out, so these values treat all sub-weekly variation as irrelevant to depletion. No epidemic outcome enters.

contacts capped at	λ	turnover at $R = 1.6$	turnover at $R = 2.5$
20	3.4–3.7	12–13%	22–24%
50	4.6–5.2	9–10%	16–18%
100	5.9–6.8	7–8%	13–14%

Table 2: Immunity factor predicted from the CoMix decomposition and the UK age matrix, without fitting to epidemic outcomes.

The immunity factor changes by only about 7% across a tenfold range of growth rates, because the switching part is small. The level depends on how the few very high contact counts are treated: it rises from 3.4–3.7 at a cap of 20 to 5.9–6.8 at 100. We lead with the cap of 20 because it limits the influence of a few reports of very large groups. That choice is not derived from transmission data, and how many of a very busy person’s contacts can transmit is unresolved.

6 The immunity factor in epidemics

US county waves. We identify 471 epidemic waves in 372 US counties of more than 50,000 people from daily deaths (New York Times, 2021), February 2020 to June 2021. For each wave, early growth gives R through a gamma generation interval of mean 5.5 days and standard deviation 2.1 days (Wallinga and Lipsitch, 2007). Cumulative deaths divided by an infection fatality rate give the infected share when the wave began, a_0 , and at the death peak, a_1 .

Suppose contacts stayed at their early-wave level. Then the wave’s growth and its infected share at the peak imply, through (2), the value

$$\hat{\lambda} = \frac{\ln R}{-\ln((1 - a_1)/(1 - a_0))}, \quad (11)$$

which we call the *implied peak index*. It equals the immunity factor only if transmission per contact and the number of contacts did not change during the wave. When contacts fell, $\hat{\lambda}$ absorbs the fall and does not measure immune selection.

Between 95% and 100% of waves in every season turned over below the homogeneous threshold. In winter 2020–21, with 341 waves and growth estimated from deaths, a constant $\hat{\lambda}$ fits at 3.5 for a fatality rate of 0.7%, 2.4 at 0.5% and 5.3 at 1%. With growth estimated from cases it is about 2.1 at 0.7%.

The slope is not identified. Grouped by growth rate, the winter waves infected nearly the same share of their susceptibles whatever their R . Growth read from deaths and from cases three

weeks earlier correlate at only 0.23 in winter, and noise in a regressor flattens any fitted slope. With case growth as an instrument the slope of log depletion on $\log \ln R$ is 0.91 in spring (90% interval 0.46 to 1.50) and 2.9 in winter (0.7 to 12.5). The county data identify the level of $\hat{\lambda}$ but not its dependence on growth, which Table 2 says is small.

Serology. Antibody surveys measure the infected share directly. Scaling each survey back to the death peak with cumulative deaths gives the infected share at the turnover and the implied peak index in Table 3.

place	survey	contacts	R	infected	$\hat{\lambda}$
England, 9 regions	Ward et al. (2021)	lockdown	2.1–3.5	0.9–5.0%	17–94
Geneva	Stringhini et al. (2020)	closures	3.3	5.1%	23
Stockholm	Castro Dopico et al. (2021)	voluntary	2.8	2.3–3.1%	33–44
Spain, spring	Pollán et al. (2020)	lockdown	1.9–4.3	0.3–4.6%	about 60
Spain, autumn	round 4	curfew	1.1–1.5	2.6% [†]	about 7
Manaus	Buss et al. (2021)	little	1.6	about 17%	2.5

Table 3: Turnovers anchored by serology; the infected share is at the turnover. [†]Share of those still susceptible after the first wave.

Where contacts were cut, by law or voluntarily, the first wave turned over with 1–5% infected and $\hat{\lambda}$ exceeds 15. These values are consistent with the observed reduction in contacts and cannot be read as immunity. Where control was weak, in Manaus and in the US county waves of winter 2020–21, $\hat{\lambda}$ is 2 to 3.5. Tkachenko et al. (2021) fit an immunity factor of about 4 to New York City and Chicago with a model that includes mitigation.

7 Discussion

Contact surveys predict an immunity factor of 3.4–3.7 at a 20-contact cap and 4.6–6.8 at larger caps. Winter county waves imply a peak index of 2.4–5.3 over fatality rates from 0.5% to 1%, 3.5 at 0.7%, and about 2.1 with growth from cases. The ranges overlap. At $R = 2.5$, $\lambda = 3.5$ puts the turnover near 23%.

On the contact side the level depends on how very high contact counts are treated and on the residual below one week. On the epidemic side it scales with the assumed fatality rate and the growth proxy. The two sets of uncertainties are of similar size.

In the panel most of the resolved heterogeneity persists. The switching part is smaller and decorrelates over about a month, so in this model a first-wave turnover is mostly a lasting reduction of the threshold with a transient on top.

The reported later attack in Manaus, 66–76%, is difficult to reconcile with a fixed-activity model of this size. Changing activity is one possible explanation. The contact distribution in Manaus and the blood-donor estimates are additional uncertainties.

The prediction can be tested out of sample. CoMix panels exist for about twenty countries. Fixing the treatment of high contact counts in advance, each panel predicts an immunity factor that the country’s serology and death curves can confirm or refute.

A Proof of Proposition 1

Let $I = (I_1, I_2)$ be the infectious people by current activity and $\Phi = a^\top I$. Early on, $\dot{I} = \beta \Phi (a \circ p) - \gamma I + Q^\top I$. Seek $I = v e^{rt}$:

$$v = \beta \Phi (r + \gamma - Q^\top)^{-1} (a \circ p). \quad (12)$$

Since $Q^\top p = 0$ and $Q^\top ((a-1) \circ p) = -\kappa (a-1) \circ p$ for a two-state chain, and $a \circ p = p + (a-1) \circ p$,

$$(r + \gamma - Q^\top)^{-1} (a \circ p) = \frac{p}{r + \gamma} + \frac{(a-1) \circ p}{r + \gamma + \kappa}. \quad (13)$$

Taking $a^\top$ of both sides, with $a^\top p = 1$ and $a^\top ((a-1) \circ p) = \sigma^2$, gives (5); R_0 is the case $r = 0$.

For the immunity factor, the susceptible deficit $s = p - S$ solves $\dot{s} = \beta \Phi (a \circ S) + Q^\top s$, and during growth $s = \beta \Phi (p/r + (a-1) \circ p/(r + \kappa))$ to first order, with overall attack $x = \beta \Phi / r$. Then $a \circ s$ has components $x(1 + \sigma^2 \rho)$ along p and $x(1 + (1 + c)\rho)$ along $(a-1) \circ p$, using $(a-1)^2 = c(a-1) + \sigma^2$. The next-generation number with susceptibles S is $\beta a^\top (\gamma - Q^\top)^{-1} (a \circ S)$, and subtracting its value at $S = p$ gives $R_e = R_0 (1 - \lambda x)$ with λ as in (6). The formulas agree with numerical solution of the model: the growth rate to four digits and λ to 0.1–0.3% (04_activity_switching.py).

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